

Integrated Modeling of Performance Monitoring with Business Artifacts

Rong Liu^a, Anil Nigam^a, Jun-Jang Jeng^a, Chian-Rou Shieh^b, Frederick Y. Wu^a

^aIBM T.J. Watson Research Center

New York, USA

{rliu, anigam, jjjeng, fywu}@us.ibm.com

^bNational Tsing Hua University

Hsinchu, Taiwan

happycircle2@yahoo.com.tw

Abstract—Real-time performance monitoring is an essential practice to enable sense-and-respond capability of enterprises. However, building a performance monitoring application is often arduous. We present a model-driven, systematic approach to performance monitoring based on an extended business artifact-centric process model. In this approach, business artifacts provide a means of tracking resource utilization in processes. Monitoring contexts and inbound event definitions are derived from the artifact model and can be configured for target performance measures. During process execution, inbound events are collected in a timely manner and fed into the monitoring contexts to calculate performance measures and detect situations. Thus, we turn performance monitoring into an integrated part of business process modeling to achieve accelerated development.

Keywords- *performance monitoring; business artifact; metric; monitoring context; performance measure*

I. INTRODUCTION

Performance management has emerged as a critical practice for enterprises to help them assess progress of business activities towards achieving business goals. A typical lifecycle of performance management involves collecting data (Sense), performing business analytics and displaying business metrics (Analyze), detecting business situations (Detect), making decisions to respond to situations (Decide), and executing decisions (Execute) [9]. Although various methods for performance management have been proposed [9],[5],[17],[22], there are challenges in performance management practices, as summarized in [4].

First, it is difficult to identify useful performance measures to capture progress towards goal attainment [4],[16]. This identification is a key to successful performance management. Studies have suggested that performance measures should be grounded in strategies [16]. However, often there is a gap between high-level business strategies and performance measures that are monitored through business operations. Instead, there is a need for a systematic approach to decompose strategic goals into operational goals that can be tracked explicitly and related back to strategic targets.

Second, performance management requires input events or data collected from business activities. Business processes are the main providers of these inputs. However, business processes are often modeled with a focus on business activities and their execution sequences using workflow technologies, without fully capturing the activity

information context. As a result, business process applications may not be able to directly provide sufficient information for performance monitoring. For example, to calculate performance measures in the category of costs and assets, information about resource usage of business activities is needed. However, typically, explicit resource modeling is not included in business process models.

In addition, there is a lack of systematic mechanisms for collecting data accurately and in a timely manner; a field study [4] confirms this observation. Often, performance management solution is a very manual process.

To address these challenges, we propose a new approach to performance measurement based on an extended concept of business artifacts [2],[18]. Business artifacts are grounded in the operational goals of a business and the results that it produces. Operational goals can be tracked back to business strategies. In this approach, the modeling of business artifacts is extended by explicitly considering resources used in transforming business artifacts through their lifecycles. The resource usage together with the information model of business artifacts provide a complete information context for calculating performance measures against operational goals. As business artifacts progress along their lifecycle, information pertinent to performance measurement can be collected automatically. We present, in this paper, a systematic approach to measuring performance based on this extended model.

Although there is abundant research on performance monitoring, most work does not take into account resources used in processes explicitly and sufficiently so that performance monitoring can be a natural and intuitive part of process modeling. The most notable work in this area is Business Activity Monitoring (BAM) [17] referred to the collection, aggregation, analysis, and presentation of real-time or near-real-time data about activities. The goal of BAM tools is to use data logged by the information system to diagnose the operational processes. An example is the IBM Cognos Performance Management Software [3]. Cognos Performance Manager extracts information from audit trails and displays this information in a graphical way (e.g., flow times, bottlenecks, utilization, etc.).

There are other real-time performance monitoring systems, for example, a semantic approach based on Web Ontology Language [22], iWISE architecture with a business rule engine [5], and policy-driven performance management [10]. The major difference between these traditional approaches and ours lies in the use of process

information including resource as the key input to the generative mechanism of the performance monitoring models. Others either do not consider processes or have a limited view of processes but this framework does.

Resource modeling is another related area. One definition is that “firm resources include all assets, capabilities, organizational processes, firm attributes, information, knowledge, etc; controlled by a firm that enable the firm to conceive of and implement strategies that improve its efficiency and effectiveness [6].” On the other hand, Amit & Schoemaker [1] suggested that the encompassing construct previously called resources can be split up into resources and capabilities. In this respect resources are tradable and non-specific to the firm, while capabilities are firm-specific and used to utilize the resources within the firm, such as implicit processes to transfer knowledge within the firm. This distinction has been widely adopted throughout the resource-based view literature. However, in this manuscript, we do not make such differentiation as we are focused the design perspective of resource-based process modeling.

A generic UML model of resources is given in [20]. Similar to our work, the purpose of modeling resources is to enable performance analysis and schedulability analysis. However, business processes are not a part of this UML model. Most commercial products of business process modeling inherently define resources only as human workers from which the business processes consume as resources. An example is IBM’s WebSphere Business Modeler which supports basic resource model analysis capabilities [12] for business analysts to perform various types of analysis e.g. role analysis, cost analysis, and availability analysis.

The rest of the paper is organized as follows. Section 2 introduces the concepts of business artifacts, resources, and performance measures, and shows that they are integral components of artifact-centric modeling of business processes. In Section 3, we present the proposed performance measurement framework with details. Section 4 illustrates this framework with an example. The implementation of this framework is sketched in Section 5. Finally, Section 6 concludes this paper with a brief description of our future work.

II. BUSINESS ARTIFACT, RESOURCE, AND PERFORMANCE MEASURES

The concept of business artifacts as the basis of constructing the operational specification has been practiced and enriched through a number of business transformation engagements [2]. Business artifacts are information entities that allow a business to define and track its operations. Let us consider the example of consumer banking. One of the basic business artifacts that a bank needs to produce and manage is an Account. Many of the bank’s operations consist of managing and reporting on the accounts that it holds. Clearly there are many other artifacts that the bank needs to maintain as well e.g. customers details, tax filings, transaction records etc. However, starting with the Account business artifact and defining its lifecycle helps to identify

other related business artifacts. In this section we discuss how the notion of business artifacts can be extended to provide a basis for business performance management.

As discussed earlier business artifacts are grounded in the operational goals of a business and the results that the business produces. Intuitively, all information that is required to analyze the progress of the business needs to be recorded on the business artifacts. An aspect of the analysis is to quantify how well the business goals are met. Another observation we make is that the operation of the business is primarily concerned with managing the resources efficiently. In this context, we define resources generally as things that are required to execute a business process [15], including tangible or intangible assets, human resources, and capabilities etc. Consequently most, if not all, key performance indicators of the business depend on amount of resources consumed, the time over which the resources are consumed, and the costs of these resources. Based on these measurements, a variety of consumption rates, elapsed times in various parts of the business operations and expenses can be computed.

Based on the belief in the primacy of resources, we will examine how the notion of resources can be incorporated as a first class entity when characterizing business artifacts as well as business operations. In our prior work on business artifacts [18], we have required that each meaningful step in the operation of the business add (or modify) some information in the business artifact – consider these as elements of results being written on to the business artifact. Here we extend the above discipline to require that information pertaining to resources also be recorded on to the business artifact. For instance sufficient information needs to be recorded on the business artifact, to enable the computation of elapsed time and resources consumed during a step in the operation. The rest of the paper will develop this argument. So our basic argument is:

- business artifacts record business-relevant results,
- efficient utilization of resources is a key goal of the business and as such is intimately reflected in the key performance indicators of the business, and
- hence resource management needs to be an integral part of artifact-centric modeling of businesses

III. REAL-TIME PERFORMANCE MONITORING

A. Generic Performance Monitoring Framework

In this section, we present a framework for real-time performance monitoring based on the extended concept of business artifacts. Performance monitoring involves gathering relevant events, correlating the events with right business contexts, and analyzing the events and contexts to gain meaningful, instant visibility to business operations. We start with a generic performance monitoring framework that is a part of Business Operations Metamodel [7] and then specify the details of each element in this framework based on the extended business artifact model. This generic framework consists of three key components: (1) *events* and *external data sources*, (2) *monitoring contexts*, and (3) *situation triggers* and *outbound events*, as shown in Fig. 1.

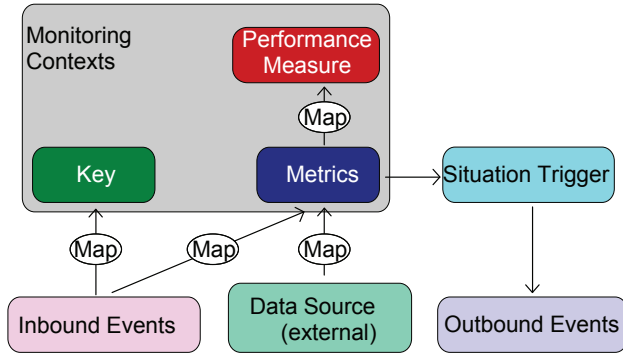


Figure 1: Generic Performance Monitoring Framework

An inbound event reports the current status of objects of interest. Each event has an event key and a payload which contains actual context data of the event. Sometimes, the status of the objects can also be obtained from external data sources. For example, the credit score of a customer may be retrieved from an external data source maintained by a credit reporting company. A monitoring context stands for an observed entity. Each monitoring context has a key, which uniquely identifies an instance of the entity, and a collection of metrics describing the status of the entity. For example, to monitor on-time rate, we can create a monitoring context for flights. The key of this monitoring context is the combination of airline, flight number and departure date. Its metrics include both scheduled and actual departure and arrival times. A monitoring context instance is created for each individual flight. Then, flight on-time rate can be calculated as an aggregation function on the metrics of the monitoring context instances. A situation trigger can also be set up on a monitoring context. For example, a trigger can be set to alert flight delays and then issue related outbound events.

However, there are several challenges in applying this generic performance monitoring framework. The key to this framework is to define events and monitoring contexts so that meaningful information can be extracted from inbound events and represented as performance measures. Most inbound events come from business processes. Usually, monitoring contexts and events are defined on a case by case basis rather than through a systematic approach. On the other hand, we explore a model-driven approach to automatically deriving inbound event definitions and monitoring contexts from business process models to achieve performance monitoring.

B. Extended Business Artifact Model

To enable performance monitoring, we extend business artifact model to incorporate resources used in business processes. Fig. 2 shows the extended business artifact UML model. In this model, business artifact class represents the information model of business artifacts, containing relevant data attributes. A business artifact is processed by business tasks, and thus, the business tasks constitute steps in its lifecycle. The highlighted classes in this model are related to resources, which extends the original business artifact

model to make business artifacts a construct of managing resources.

Various resources, such as physical resources, human resources, and intangible assets, may be needed to execute business tasks in the lifecycle of a business artifact. For example, for running a restaurant, necessary resources can be food, tables, waiters, and chefs. As described in the Resource class in Fig. 2, a resource can be consumable (e.g. food) or unconsumable (e.g. tables, waiters). A resource may have fixed capacity or its availability can be scheduled. Note that in Fig. 2, Capacity is modeled as a complex type and contains attributes describing resource scheduling information and resource quantity if applicable. Thus a resource can be requested or used for a period of time. In addition, a resource may play multiple roles or provide more than services. For example, a waiter may act as a cashier also. Each resource or resource service can be associated with a cost profile describing the cost of consuming the resource.

Business tasks request necessary resources for task execution. Inspired by [19], we model the requested resources as resource demand as shown in Fig. 2. The actual resource usage of business tasks is recorded for each resource demand. Resource managers [20] often act as resource brokers to allocate resources to demand based on resource allocation policies. In addition, resource managers may update resource status and maintain resource usage records. The details of resource managers and resource allocation policies are out of the scope of this paper. Any typical resource manager, for instance, the one described in [20] can be integrated with our model.

With this extended business artifact model we can track the following information of a business artifact: *attribute data, business tasks in its lifecycle and the task execution logs, resource demand and resource usage* of the tasks. These pieces constitute *the information content of the business artifact*. Next, we introduce performance monitoring using the extended business artifact model.

C. Performance Monitoring

Our performance monitoring model is driven by performance measures. Performance measures need to be aligned with business strategies, and the ultimate goal of performance monitoring is to stimulate actions so that the business strategies can be realized [16]. An actionable business strategy typically defines objectives, metrics to evaluate the success of the strategy, and initiatives leading to the success. For instance, a pharmaceutical company with a strategic alliance defined a business strategy where the objective is to “develop transparent cost drivers”, metrics include “quality and risk assessment score of development plan” and “trust and transparency survey score”, and initiatives are set to “create a new development plan process” [8]. In this example of business strategy, performance measures are explicitly specified as metrics and initiatives set operational goals which may require appropriate business artifacts in support of the operational goals. For instance, in this case, development plan may qualify as a business artifact to start with.

Fig. 3 shows our methodology for creating performance monitoring models. High-level performance measures can be decomposed into low-level and more operational measures using techniques such as Business System Design Decomposition and Analytical Hierarchy Process [21]. We argued that most performance measures can be represented as functions of resources and artifact information. Hence, we keep decomposing business measures until the low-level measures can be computed based on resource and artifact information content.

Interestingly, we can consider that each business artifact has a default monitoring context with associated inbound events. The key of this monitoring context is simply the artifact ID. The artifact information content provides the metrics of the monitoring context, which includes, as shown in Fig. 3, artifact attributes, business tasks (including execution logs) in the artifact lifecycle, resource usage, and resource demand. Inbound events are the “lifeblood” that affects changes to the metrics. For a business artifact, an important principle is that when business tasks in its lifecycle are executed, business value is added through the updates made to its data. Hence, changes on artifact information are always caused by task execution. Therefore, we can identify the following types of events: (1) events that start, complete, suspend, resume, or cancel business tasks, (2) events that update resource demand of business tasks, and (3) events that begin (or stop) using resources. As shown in TABLE I, each event has a key and a payload. For these events, the key is the artifact ID and the payload is the artifact information. Note that the key of an event is for the purpose of matching the event with a monitoring context, not necessarily a unique ID.

TABLE I describes the default monitoring context of a business artifact. Certainly, this default monitoring context can be configured on demand. For example, we can select a portion of the artifact information as the metrics of the configured monitoring context. Accordingly, only some types of events will change the selected metrics.

Similarly, each resource also has a default monitoring context, as shown in TABLE I. The key to this monitoring context can be the unique resource ID or resource service ID if the resource provides multiple services. The metrics of this monitoring context come from resource information including the attributes of Resource, Resource Service, and Cost Profile classes. Inbound events are those that update the resource information, for example, resource status, capacity and cost profile. The key to the events is the resource ID (or resource service ID) and the payload is the resource information.

TABLE I describes the default monitoring contexts of business artifacts and resources. A default monitoring context can be configured in two aspects: metrics and inbound events. Only metrics needed for the computation of target performance measures are selected. Accordingly, only events that update these selected metrics are selected. When a selected event arrives, the monitoring context first checks if there is any monitoring context instance with a key matching the event key. Then three possible event processing strategies are taken as shown in TABLE II. For example, the completeTask of “Create Guest Check” can be associated with “Create(Guest Check)” strategy. This strategy creates a new instance of Guest Check monitoring context when this event happens.

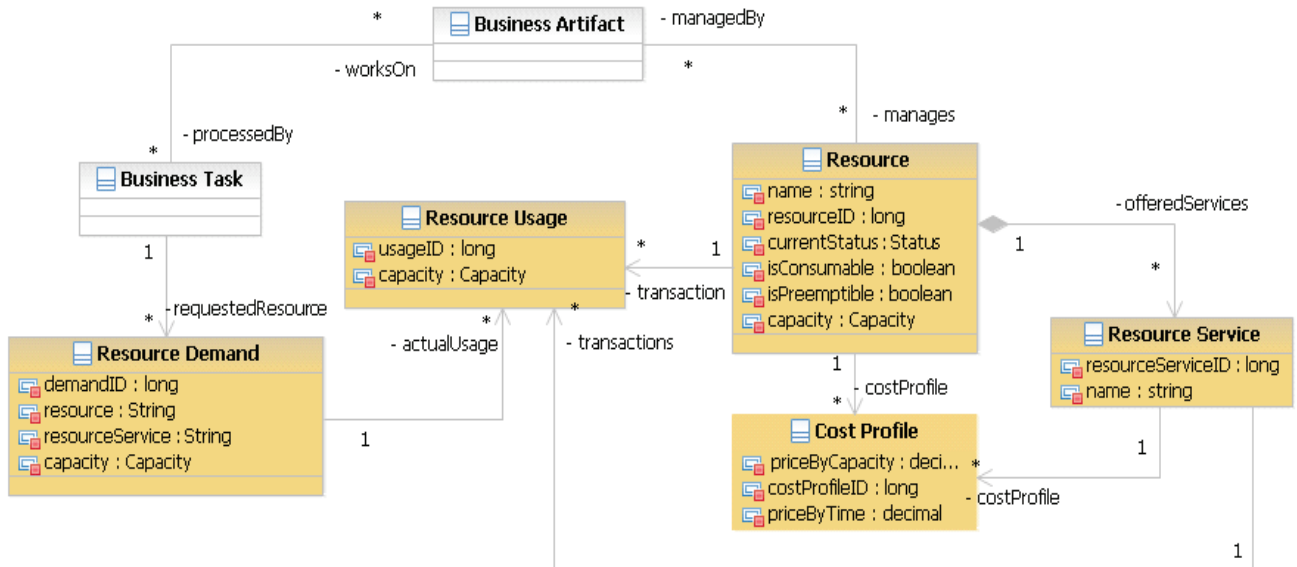
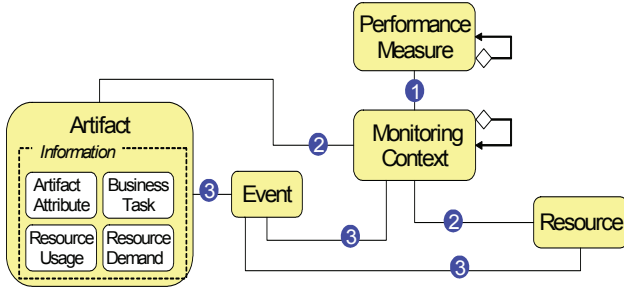


Figure 2: Extended Business Artifact Model



- (1) Decompose performance measures so that they can be mapped to the information of some business artifacts and/or resources
- (2) Configure the metrics in the default monitoring contexts of the business artifacts and resources to match with the decomposed performance measures
- (3) Configure the events in the default monitoring contexts of the selected business artifacts and resources to match with the configured metrics

Figure 3: Performance Monitoring Methodology

Next, we use an example to illustrate the business artifact-centric performance monitoring model.

IV. BUSINESS ARTIFACT, RESOURCE, AND PERFORMANCE MEASURES

An example restaurant business process is described in [18], as shown in Fig. 4. This business process starts with a guest check created when a guest arrives (Task: Create Guest Check). Ordered food items are added to the guest check and kitchen orders are created (Task: Add Items to Guest Check). Then, food is prepared by the kitchen orders (Task: Prepare Items). Finally, the guest makes a payment (Task: Tender Guest Check). In this simple business process, two business artifacts are identified: *Guest Check* and *Kitchen Order*.

For each task, we can easily identify resources required for task execution. For example, when creating a Guest Check, a waiter and a table are assigned. In fact, in a typical guest check, we can also find important resource information, such as waiter ID and table ID, as attributes of Guest Check information model, as shown in Fig. 5. In

general, through the extended business artifact model, we can track the resource used in a business process through business artifact data attributes, resource demand or resource usage. This confirms that business artifacts act as a construct for managing resources.

TABLE III lists the resource demand of the business tasks in this process. For instance, when creating a Guest Check, required resources are a table, a waiter and menu. These resources are not consumable. They are locked for use for some period of time during the task execution and will return to a free state after the task is complete. Hence, resource usage entries may be produced to record when a resource is assigned for use and when the resource is released. For example, when completing the task Create Guest Check, a table is assigned for use, and this table is released when task Tender Guest Check is executed. However, some resources are consumable. For example, food material is consumed by Prepare Items task.

There are a number of performance measures for monitoring a restaurant business, for example, *Number of Customers*, *Seat Utilization*, *Revenue per Available Seat Hour*, *Food Cost Percentage*, *Kitchen Labor Percentage*, and *Total Labor Cost Percentage* etc. Some metrics are “point-in-time” (for instance, Seat Utilization described here) and others are “time period” metrics. Next, we design real-time monitoring models for Seat Utilization to illustrate our approach.

Following the methodology shown in Fig. 3, we first decompose performance measure Seat Utilization until it can be mapped to some artifact information. In the restaurant business process, Seat Utilization is the percentage of tables occupied by guests. Thus, we need to count the number of tables in service. According to the Guest Check business artifact shown in Fig.5, each Guest Check has a Table ID. Then, we can get the number of occupied tables by counting the number of active Guest Check instances. Therefore, this performance measure can be directly mapped to Guest Check business artifact information.

TABLE I. DEFAULT MONITORING CONTEXT

Default Monitoring Context Type	Default Monitoring Contexts				
	Key	Metrics	Inbound Events		
			Name	Key	Payload
Business Artifact	Artifact ID	Artifact information content	startTask, completeTask, suspendTask, resumeTask, cancelTask, updateResourceDemand, beginResourceUse, stopResourceUse	Artifact ID	Artifact Information
Resource	Resource ID / Resource Service ID	Resource, Resource Service, Cost Profile	updateResource	Resource ID	Resource Information

TABLE II. INBOUND EVENT PROCESSING STRATEGIES

Strategy	Semantics
<i>Create(x)</i> <i>x: business artifact or resource</i>	When no matching instance of monitoring context <i>x</i> is found, create a new monitoring context instance, and populate the metrics of the created instance using the event payload
<i>Update(x)</i>	If there is a matching instance of monitoring context <i>x</i> , update the metrics of this instance using the event payload. Otherwise, this strategy is equivalent to <i>Create</i> strategy
<i>Destroy(x)</i>	destroy matched instances of monitoring context <i>x</i>

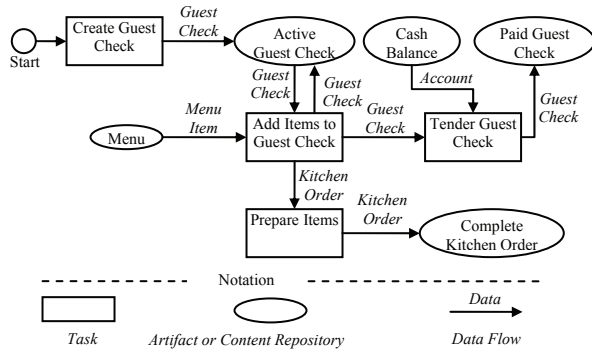


Figure 4: Business Artifact-Centric Model for Restaurant Business Process

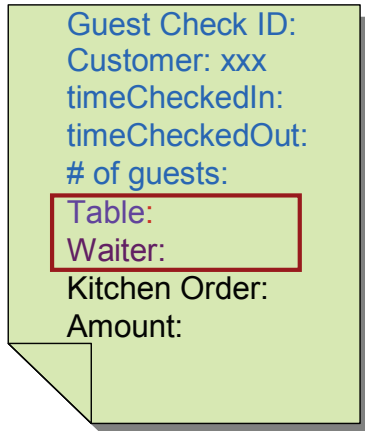


Figure 5: Guest Check Business Artifact

Then we can configure the metrics in the default monitoring context of Guest Check business artifact. The information content of the Guest Check business artifact contains many data fields, but to monitor Seat Utilization, only attribute Table ID is sufficient. With metrics selected, the next step is to configure inbound events using the strategies shown in TABLE II.

For example, when “CreateGuestCheck” completeTask event happens, it is configured with “Create” strategy so

that a new monitoring context instance is created with the Guest Check ID as the key. When the instance is created, the payload from the event is populated to the metrics of the monitoring context. For instance, the table ID and the number of guests are populated in the created monitoring context instance, as shown in Fig. 6. When “Tender Guest Check” completeTask occurs, this indicates a Guest Check has completed its lifecycle and it is not active any more. Therefore, we can use “Destroy” strategy to delete the monitoring context instances matching with this event, as shown in Fig. 6.

When new values are populated into the metrics of a monitoring context instance, performance measures are calculated (or recalculated). For example, a counter is designed to count Guest Check monitoring context instances, as shown in Fig. 6, to get the number of occupied tables and then calculate real-time Seat Utilization of this restaurant. Meanwhile, a situation can be defined once the counter reaches the total number of tables in this restaurant. This situation can trigger an outbound event to warn that the restaurant is full.

More complicated monitoring contexts can be defined by correlating multiple business artifact monitoring contexts or by correlating business artifact monitoring context with resource monitoring context. For example, to monitor Food Cost Percentage per Guest, we need to use two business artifacts, Guest Check and Kitchen Order, assuming that the cost of each order item can be found by the Kitchen Order business artifact. These two business artifacts can be correlated at Kitchen Order ID, since the Guest Check business artifact has Kitchen Order ID as its data attribute. Therefore, this performance measure can be calculated by two selected metrics: amount from Guest Check, and cost from Kitchen Order. The events matching this monitoring context include completeTask event of “Create Guest Check”, completeTask of “Add Items to Guest Check”, completeTask of “Prepare Items”, and completeTask of “Tender Guest Check. As shown in Fig. 7, an instance of this monitoring context contains an instance of Guest Check monitoring context and perhaps multiple instances of Kitchen Order monitoring context, because a guest may require multiple Kitchen Orders. When completeTask event of “Create Guest Check” arrives, an instance of Guest Check monitoring context is created using Create event processing strategy. Task “Add Items to Guest Check” works on two business artifacts: creating a new Kitchen Order instance and updating the Kitchen Order ID in the Guest Check instance. The completeTask event of this task requires both Create and Update event processing strategies. The Create strategy on business artifact Kitchen Order creates a new instance of Kitchen Order monitoring context and the Update strategy on Guest Check adds value to the Kitchen Order ID metric in Guest Check monitoring context. When “Prepare Items” completes, the completeTask event fills in the Cost Metric in Kitchen Order monitoring context using Update strategy. Finally, when “Tender Guest Check” completeTask event occurs, the Amount in the Guest Check monitoring context is updated.

TABLE III. RESOURCE DEMAND

Task	Resource Demand
Create Guest Check	Table, Waiter, Menu
Add Items to Guest Check	Waiter
Tender Guest Check	Cashier
Prepare Items	Chef, Food material, Kitchen

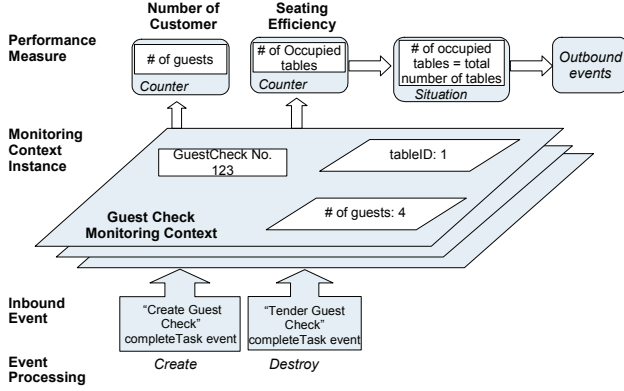


Figure 6: Real-time Monitoring Model for Seat Utilization

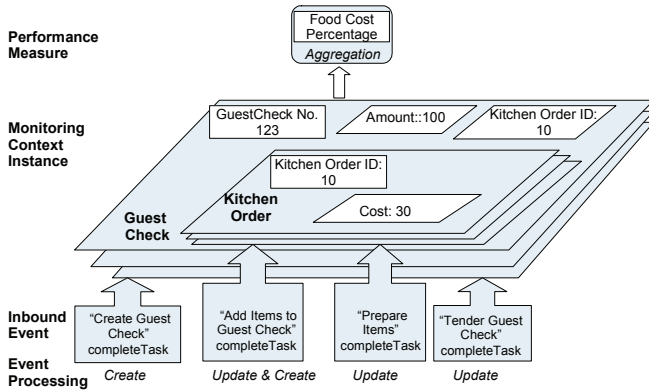


Figure 7: Real-time Monitoring Model for Food Cost Percentage

As another example, to monitor Revenue per Available Seat Hour, we can create a monitoring context containing Guest Check business artifact and Table resource, which are correlated by Table ID, the key for table resource and a data attribute in Guest Check business artifact. The metrics of this monitoring context include amount, check-in and check-out times from Guest Check, capacity of Table, and Table availability. Inbound events of this monitoring context are updateResource event and completeTask event of “Tender Guest Check”. When the status of Table changes (i.e. an updateResource event occurs), either a new Table resource monitoring context instance is created or a matching instance is updated (using Update strategy). To calculate revenue, Guest Check monitoring instances are needed. These instances can be created by completeTask events of “Tender Guest Check” (Using Create strategy), since the payload of these events has “amount” information. Then, the Guest Check monitoring context instances are

correlated with Table resource monitoring context instances on Table ID.

V. IMPLEMENTATION

In this section, we sketch the implementation of our performance monitoring model. The implementation architecture is shown in Fig. 8. We take a model-driven approach to automatically or semi-automatically generate an implementation of business monitoring. As the figure shows, first an artifact-centric process model, which can be designed using Webspere Business Modeler [12], is transformed into an implementation model, for example, as BPEL flows or business state machines [11]. The implementation model can be executed using Websphere Process Server [13]. During runtime, process and resource events (see TABLE I) are emitted. Meanwhile, a performance monitoring model consisting of default monitoring contexts is generated from the process model in accord with the Observation meta model specified in Business Operation MetaModel [7]. The default monitoring contexts can be configured using Websphere Business Monitor Model Editor [14]. Finally, the configured monitoring contexts are executed using Websphere Business Monitor to create monitoring context instances when inbound events are received. As a result, the real-time performance of the process can be monitored and dashboards or analytics can be created from these instances.

VI. CONCLUSION

In this paper, we have presented a model-driven approach to real-time performance monitoring based on an extended business artifact model. In the extended model, business artifacts provide a means for tracking resource and measuring resource utilization in business processes. Thus, performance monitoring model targeting at specific performance measures can be automatically derived from the extended model. Specifically, each business artifact can be treated as a monitoring context with artifact information as metrics. Inbound events mainly originate from business tasks in the lifecycle of business artifacts. Hence, while processes are executed, inbound events can be systematically collected in a timely manner and automatically match with business artifact monitoring contexts. In general, a performance monitoring model can be easily configured by selecting appropriate business artifact or resource monitoring contexts, determining metrics in the contexts, choosing inbound events and event processing strategies, and correlating the monitoring contexts if necessary.

We expect our future work to design a tool which can generate monitor contexts from artifact-centric process definitions and target performance measures. Also, we plan to apply our performance monitoring approach to real business engagements. This approach will be enriched with features to enable business analytics to offer predictive ability instead of retrospective views of performance.

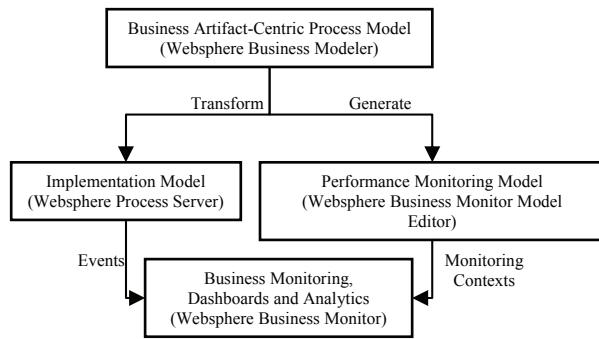


Figure 8: Implementation Architecture of Performance Monitoring Model

REFERENCES

- [1] Amit, R.; Schoemaker, P.J.H. (1993), Strategic assets and organizational rent. *Strategic Management Journal*, 14, (1), pp.33–46.
- [2] Bhattacharya, K., Caswell, N. S., Kumaran, S., Nigam, A. and Wu, F. Y., Artifact-centered Operational Modeling: Lessons from Customer Engagements, *IBM Systems Journal*, 46(4):703-721, 2007.
- [3] Cognos Business Intelligence and Financial Performance Management <http://www-01.ibm.com/software/data/cognos/>
- [4] Corea, S. and Watters, A. Challenges in Business Performance Measurement: The Case of a Corporate IT Function, In *Proceedings of the 5th International Conference on Business Process Management (BPM 2007)*, PP 16-31, 2007.
- [5] Costello, C. and Molloy, O. Building a Process Performance Model for Business Activity Monitoring, *Information Systems Development Challenges in Practice, Theory, and Education*, Volume 1, pp 230-241, 2009.
- [6] Daft, L.R., (1983), *Organizational Theory and Designs*, West Pub. Co. , St. Paul.
- [7] Frank, J. H. and Stefanos, G. G., *Business Operations Metamodel (BOM)*, Version 2005.
- [8] Kaplan, R., norton, D. P., and Rugelsjoen, B. "Managing Alliance with the Balance Scorecard," *Harvard Business Review*, pp. 114-120, January-February, 2010.
- [9] Jeng, J. J. Service-Oriented Business Performance Management for Real-Time Enterprise, *The 8th IEEE International Conference on E-Commerce Technology and The 3rd IEEE International Conference on Enterprise Computing, E-Commerce, and E-Services (CEC/EEE'06)*, pp 28, 2006.
- [10] Jeng, J. J. Chang, H., Bhaskaran, K: Policy Driven Business Performance Management. *DSOM 2004*: 52-63.
- [11] Kumaran, S., Nandi, P., Heath, T., Bhaskaran, K. and Das, R. ADoc-oriented program-ming, in *Symposium on Applications and the Internet (SAINT)*, pp. 334–343, 2003.
- [12] IBM Websphere Business Modeler, <http://www-01.ibm.com/software/integration/wbimodeler>
- [13] IBM Websphere Process Server, <http://www-01.ibm.com/software/integration/wps/>
- [14] IBM Websphere Business Monitor, <http://www-01.ibm.com/software/integration/wbimonitor/>
- [15] McDavid, D. W., A Standard for Business architecture Description, *IBM Systems Journal*, 38(1):12-31, 1999.
- [16] Neely, A., Richards, H., Mills, J., Platts, K. and Bourne, M. Designing performance measures: a structured approach. *International Journal of Operations & Production Management* 17, 1131-1152, 1997.
- [17] Nesamoney, D. BAM: Event-Driven Business Intelligence for the Real-Time Enterprise, *Information Management Magazine*, March 2004.
- [18] Nigam, A. and Caswell, N. S., Business Artifacts: An Approach to Operational Specification, *IBM Systems Journal*, 42(3):428-445, 2003.
- [19] Object Management Group. UML Profile for Schedulability, Performance and Time Specification, March 2002
- [20] Selic, B. A Generic Framework for modeling Resources with UML, *Computer* 33(6):64-69, 2000.
- [21] Taticchi, P., Tonelli, F. and Cagnazzo, L. A decomposition and hierarchical approach for business performance measurement and management, *Measuring Business Excellence* 13(4):47-57, 2009.
- [22] Thomas, M., Redmond, R., Yoon, R. and Singh, R. A Semantic Approach to Monitor Business Process Performance. *Communication of the ACM*, 48(12): 55-59. 2005